Skills Summary

Approximating Irrational Roots on a Number Line

Use this reference while completing your worksheet or when studying.

Perfect squares to know: $1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169$.

The one rule everything uses: for positive numbers, squaring keeps the order. So $p < q$ exactly when $p^2 < q^2$ — which means you can compare roots without ever computing one, just by squaring the number you are comparing it to.

1. Trap the root between two whole numbers

Find the perfect square just below the radicand and the one just above it. If $p^2 < N < q^2$ with $q = p + 1$, then $p < \sqrt{N} < q$. Every later estimate lives inside that gap.

Example: Between which two consecutive whole numbers is $\sqrt{30}$?
Step 1. 30 is not a perfect square, so the root is irrational — trap it between the perfect squares on either side.
Step 2. The perfect squares around 30 are 25 and 36, whose roots are 5 and 6, and $25 < 30 < 36$.
$\Rightarrow \quad \mathbf{5 < \sqrt{30} < 6}$

Try it: Between which two consecutive whole numbers is $\sqrt{90}$?

check: $\mathbf{9 < \sqrt{90} < 10}$

2. Squeeze it to the nearest tenth

Inside the bracket, test tenths by *squaring* them — never by guessing. Find the two tenths whose squares straddle the radicand, then take whichever square is closer to it. Write the answer with $\approx$, never $=$: the decimal never ends.

Example: Estimate $\sqrt{52}$ to the nearest tenth.
Step 1. The bracket is 7 to 8, so the answer is 7-point-something — test tenths by squaring them.
Step 2. $7.2^2 = 51.84$ is just under 52, and $7.3^2 = 53.29$ is well over, so 52 is much nearer the lower tenth.
$\Rightarrow \quad \sqrt{52} \approx \mathbf{7.2}$

Try it: Estimate $\sqrt{68}$ to the nearest tenth.

check: **8.2**

3. Plot it on the number line

Label the line with the tenths inside your bracket, then put the dot at the tenth your squaring chose — nudged left if the radicand was below that tenth's square, right if it was above. The dot's position, not the label, is what shows you understand the size of the number.

Example: Plot $\sqrt{23}$ on a line marked in tenths.
Step 1. First find which tenth it belongs to, then read the position off the line.
Step 2. $4.7^2 = 22.09$ and $4.8^2 = 23.04$, so 23 sits just below the upper tenth's square and the dot goes a hair left of 4.8.
$\Rightarrow \quad \sqrt{23} \approx \mathbf{4.8}$

Try it: Which tenth does $\sqrt{33}$ plot at?

check: **5.7**

4. Order roots together with rational numbers

Do not estimate the root and compare decimals — square the *rational* number instead and compare it with the radicand. It is exact, and it is faster: $\sqrt{N}$ versus d becomes N versus d^2.

Example: Which is larger, $\sqrt{11}$ or 3.4?
Step 1. One side is irrational, so square the rational side and compare radicands — that decides it exactly.
Step 2. $3.4^2 = 11.56$, and $11 < 11.56$, so the root is the smaller one.
$\Rightarrow \quad \sqrt{11} < 3.4$

Try it: Write $<$ or $>$ between $\sqrt{27}$ and 5.1.

check: $\sqrt{27} > 5.1$